

1 Part 1: Executive Summary of a Position Paper

1.1 Purpose and Audience

Mental Workload (MWL) is widely used in **Human-Computer Interaction (HCI)** to evaluate how mentally demanding an interaction is for a user. However, the concept is not defined consistently across disciplines, and *NASA-TLX* has become the de facto standard despite continuing theoretical and methodological concerns [2]. Babaei et al. [2] therefore ask three broad questions: what is MWL, how should it be measured, and are current MWL measures valid for HCI? Their contribution combines a theoretical review, a systematic review of 75 CHI papers, and a validation experiment comparing *NASA-TLX/R-TLX* with the **Multiple Resources Questionnaire (MRQ)**. The main audience is HCI researchers and practitioners who use workload measures to evaluate digital systems and need their measurement approach to match their theoretical purpose. This is also relevant to data science because unreliable or poorly defined measurement constructs can affect the quality of the data used to evaluate and compare interactive systems.

1.2 Main Challenges

The first challenge is conceptual ambiguity. MWL has been defined from task-centred, operator-centred, and performance-centred perspectives, and there is still debate over whether it should be treated as a hypothetical construct or an intervening variable. It is also unclear whether MWL should be represented as one overall score or as multiple cognitive resources. This lack of a shared definition makes results difficult to compare across studies [2].

The second challenge is how *NASA-TLX* is used in practice. The review found that researchers sometimes used *NASA-TLX* for different constructs, modified its subscales or scoring, or used it without clearly justifying why MWL was relevant. Mental Demand is also sometimes confused with Mental Workload even though they are not equivalent. These practices can reduce the validity and comparability of results [2].

The validation experiment provided further evidence of the problem. *NASA-TLX/R-TLX* and *MRQ* showed only a small positive relationship, making convergent validity questionable. Both questionnaires also failed to show a significant workload difference between objectively different task conditions, raising concerns about sensitivity. Their reliability was acceptable, but neither questionnaire strongly predicted performance [2]. The authors therefore warn against treating one MWL score as a complete explanation of user performance.

1.3 Opportunities and Recommendations

The paper recommends defining the intended MWL construct before selecting a measurement tool, justifying why MWL is relevant, and reporting administration and scoring transparently. A multivariate, resource-based approach may also be more useful for complex HCI tasks. *MRQ* is potentially more actionable for design because its subscales identify specific cognitive resources, but it is not yet proven to be a universally better replacement for *NASA-TLX* and still requires validation. Using multiple measures, such as subjective workload, performance, errors, and usability, may provide a more complete picture [2].

1.4 Case Study: Hospital / Nurse Dashboard

A practical example is the design of a nurse dashboard integrated with an Electronic Health Record (EHR). Hospital dashboards visually organise clinical and operational information, while quality dashboards can help nurses monitor safety information and make informed decisions [5, 8]. This is an important design problem because poorly organised EHR information can increase nurses' cognitive work and make it harder to maintain an overview of the patient [7].

Consider a hospital comparing its existing EHR interface with a redesigned nurse dashboard for patient monitoring. The team wants to know whether the new design reduces MWL. This is a realistic scenario. Fuller et al. [4] evaluated an inpatient safety dashboard and found improvements in efficiency without increasing cognitive burden, while Taylor-Vaughan et al. [6] found that a spatially integrated EHR interface reduced perceived workload and improved task performance for ICU nurses.

The first challenge is deciding what MWL means in this study. The researchers need to specify whether they are measuring task demand, the nurse's mental effort, or workload in relation to performance. Without this definition, a lower score is difficult to interpret.

The second challenge is instrument choice. *NASA-TLX* can provide an overall workload score and subscale ratings, but these may not identify which specific cognitive resources are overloaded. For design purposes, *MRQ* may provide more actionable information by separating resource demands. However, based on Babaei et al. [2], *MRQ* should not automatically be assumed to be more valid.

The third challenge is sensitivity. Even if the redesigned dashboard improves completion time or decision accuracy, a subjective MWL questionnaire may fail to detect the difference. A stronger evaluation would therefore combine a clearly defined MWL measure with task completion time, error rate, decision accuracy, and usability [2]. In this case, MWL becomes not only an evaluation score but part of a broader design process.

1.5 Conclusion

The paper does not argue that MWL or *NASA-TLX* should be abandoned. Instead, it shows that HCI researchers need clearer definitions, better-justified instruments, stronger validation, and more careful interpretation [2]. The nurse-dashboard case illustrates why these issues matter in practice: a workload score is useful only when researchers are clear about what it measures and combine it with evidence that supports the design decision.

2 Part 2: Deliberation on Task 1

2.1 Context

For Individual Task 1 Part 1, I used two public government datasets: the **ABARES** Australian Crop Report [1] and the Bureau of Meteorology Climate Data Online [3]. The aim was to develop a model that examined the relationship between weather conditions and Victorian wheat yield. Weather data came from **Horsham**, **Mildura**, and **Ouyen**, and I compared Linear Regression with Random Forest.

2.2 Unbiased Evaluation: Cross-Validation and Sampling

I used three weather stations to represent major wheat-growing areas in western and north-western Victoria. For each station, I collected rainfall, mean maximum temperature, and mean minimum temperature [3]. I restricted the weather data to the seven-month growing season from April to October. From 114 station-years, 105 contained all seven months and were retained, while incomplete observations were removed. The three stations were then averaged into one yearly value representing the cropping belt.

For the crop dataset, I calculated Victorian wheat yield from production and sown area [1].

Both models were evaluated using *5-fold cross-validation*. Each fold used approximately 80% of the observations for training and 20% for testing, with every observation used as test data once. This was more appropriate than one

train-test split because the final dataset contained only 37 yearly observations. Testing the model across several subsets reduced the chance that the reported performance depended on one unusually easy or difficult split.

However, the evaluation was not fully unbiased. The small dataset meant that each test fold contained only a few observations, while three stations could not represent all climatic conditions across Victoria. The most important issue was information leakage during detrending: the long-term yield trend was calculated from the full dataset before cross-validation, meaning held-out years influenced the trend estimate. A stricter approach would fit the trend separately inside each training fold.

The notebook also used shuffled *K-fold cross-validation* even though the observations were annual and time-ordered. This allows later years to appear in training data when an earlier year is being tested, which is less representative of a real forecasting scenario. A time-aware approach, such as rolling-origin validation or `TimeSeriesSplit`, would provide a stricter evaluation for future forecasting.

2.3 Security Risk

The model is still experimental and should not be used directly for high-stakes agricultural decisions. The small sample and missing predictors mean that predictions can still be inaccurate. In a real deployment, unauthorised access to the model or data pipeline could also allow someone to alter weather data, model parameters, or prediction code and produce misleading forecasts. Access control, data-integrity checks, version control, and secure storage would therefore be important.

2.4 Privacy Risk

The current analysis has low privacy risk because the **ABARES** and **BOM** datasets are public government datasets [1, 3] and do not contain personal information about individual farmers. Privacy concerns would increase if the model were later applied to farm-level information such as property locations, production records, financial information, or individual management practices. Those data would require appropriate consent, restricted access, and secure handling.

2.5 Ethical Risks and Bias

The analysis found a relationship between weather and Victorian wheat yield, especially rainfall, but weather does not explain the full variation in yield. The model excludes important factors such as soil type, pests and disease, rainfall timing, and individual farm-management decisions. The best model still left approximately 51.6% of yield variation unexplained.

There is therefore an ethical risk if users treat the prediction as a complete representation of reality. Farmers outside the western and north-western areas represented by the three weather stations may also receive less reliable predictions. A model can be statistically accurate on the available sample while still being unsuitable or unfair for groups that are poorly represented.

Even with more data and a more advanced model, I would not use the system to make agricultural decisions automatically. Its role should be to support the judgement of farmers, agronomists, and other domain experts by showing what may happen under similar conditions, not to replace their decision-making.

2.6 Conclusion

Overall, 5-fold cross-validation made the evaluation more reliable than a single train-test split, but it did not remove all bias. The small dataset, limited geographical coverage, shuffled evaluation of time-ordered observations, and leakage

during detrending all limit how confidently the results can be generalised. The current public datasets create low privacy risk, but deployment with farm-level data would introduce additional security, privacy, and ethical concerns. The model should therefore be treated as a decision-support tool rather than a decision-maker, especially while substantial yield variation remains unexplained.

3 Part 3: Peer Review

3.1 Presentation Reviewed

Topic: Predicting Delivery Risk & Customer Churn

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The presentation focused on two main business problems: late deliveries and silent customer churn. Late deliveries can reduce customer trust, while silent churn happens when customers gradually stop purchasing without directly informing the company.

The analysis used more than **96,000** records containing information such as prices, freight, product categories, and delivery timing, covering nearly **6,000** customers across approximately two years.

The first finding focused on delivery risk. Predicting late deliveries is difficult because only around **8%** of orders were late, and delays can be influenced by many external factors. However, the better-performing model was able to identify around 4 out of 10 late deliveries in advance.

The second finding focused on customer churn. By using purchasing signals such as how often customers bought, how much they spent, and how recently they purchased, the model was able to predict approximately **70%** of customer churn. The main recommendation was to begin using purchase-history predictions to identify customers at risk of churn while continuing to collect more detailed delivery data.

3.2 Soundness and Significance of the Discussion

In my opinion, the approach was relatively simple but still interesting. The conclusions were supported by the results and visualised data. The analysis was also significant because e-commerce companies, and other industries that rely on repeat customers, could benefit from models that identify customers who may churn. The delivery-risk analysis was useful because it acknowledged the limitations of the available data rather than overstating model performance.

3.3 Executive-Level Insights

The insights were simple and direct, making them suitable for an executive-level audience. The presentation avoided unnecessary technical terminology and focused on what the results meant for the business. The recommendation to use the churn model now while continuing to improve the delivery-risk model also gave the audience a clear direction.

3.4 Presentation Quality

One aspect I particularly liked was the visualisation. The graphs were well designed, engaging, and represented the data clearly. They helped communicate the findings without requiring the audience to understand the technical details behind the models.

3.5 Suggestions for Improvement

My main suggestion would be to improve the narration and audio clarity. I had to replay several parts because the speaker's voice was sometimes difficult to hear. Some findings, especially the correlations and model results, could also use slightly more explanation so the audience can understand their meaning without replaying the video.

Overall, I enjoyed the presentation. The visualisations were strong, the business problem was relevant, and the recommendations were practical.

Generative AI Attribution and Declaration

I certify that the submission of this assignment is incomplete without the submission of the completed **Condition 3 Bounded Process AI Declaration Form** (templates available in the specification of the assignment on Canvas).

Acknowledgments

I acknowledge the Traditional Custodians of the lands on which I study and work, and pay my respects to their Elders past and present.

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Development Evidence

The GitHub repository contains my drafting notes, intermediate versions, revisions, and evidence of changes made during the development of this assignment. It also documents issues identified during the process, including changes to my interpretation of the position paper, reference formatting, methodological reflection, and LaTeX formatting.

<https://github.com/s4078157/Case-Study-Data-Science-Assignment-2->